

INTERNATIONAL STANDARD

**Industrial communication networks – Fieldbus specifications –
Part 3-11: Data-link layer service definition – Type 11 elements**



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Part 3-11: Data-link layer service definition – Type 11 elements**

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**INDUSTRIAL COMMUNICATION NETWORKS –
 FIELDBUS SPECIFICATIONS –**
Part 3-11: Data-link layer service definition – Type 11 elements

FOREWORD

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International Standard IEC 61158-3-11 has been prepared by subcommittee 65C: Industrial networks, of IEC technical committee 65: Industrial-process measurement, control and automation.

This first edition and its companion parts of the IEC 61158-3 subseries cancel and replace IEC 61158-3:2003. This edition of this part constitutes a technical addition. This part and its Type 11 companion parts also replaces IEC/PAS 62406, published in 2005.

This edition includes the following significant changes with respect to the previous edition:

- a) deletion of the former Type 6 fieldbus, and the placeholder for a Type 5 fieldbus data-link layer, for lack of market relevance;
- b) addition of new types of fieldbuses;
- c) division of this part into multiple parts numbered 3-1, 3-2, ..., 3-19.

The text of this standard is based on the following documents:

FDIS	Report on voting
65C/473/FDIS	65C/484/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with ISO/IEC Directives, Part 2.

The committee has decided that the contents of this publication will remain unchanged until the maintenance result date indicated on the IEC web site under <http://webstore.iec.ch> in the data related to the specific publication. At this date, the publication will be:

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

NOTE The revision of this standard will be synchronized with the other parts of the IEC 61158 series.

The list of all the parts of the IEC 61158 series, under the general title *Industrial communication networks – Fieldbus specifications*, can be found on the IEC web site.

INTRODUCTION

This part of IEC 61158 is one of a series produced to facilitate the interconnection of automation system components. It is related to other standards in the set as defined by the “three-layer” fieldbus reference model described in IEC/TR 61158-1.

Throughout the set of fieldbus standards, the term “service” refers to the abstract capability provided by one layer of the OSI Basic Reference Model to the layer immediately above. Thus, the data-link layer service defined in this standard is a conceptual architectural service, independent of administrative and implementation divisions.

INDUSTRIAL COMMUNICATION NETWORKS – FIELDBUS SPECIFICATIONS –

Part 3-11: Data-link layer service definition – Type 11 elements

1 Scope

1.1 Overview

This part of IEC 61158 provides common elements for basic time-critical messaging communications between devices in an automation environment. The term “time-critical” is used to represent the presence of a time-window, within which one or more specified actions are required to be completed with some defined level of certainty. Failure to complete specified actions within the time window risks failure of the applications requesting the actions, with attendant risk to equipment, plant and possibly human life.

This standard defines in an abstract way the externally visible service provided by the Type 11 fieldbus data-link layer in terms of

- a) the primitive actions and events of the service;
- b) the parameters associated with each primitive action and event, and the form which they take; and
- c) the interrelationship between these actions and events, and their valid sequences.

The purpose of this standard is to define the services provided to

- the Type 11 fieldbus application layer at the boundary between the application and data-link layers of the fieldbus reference model, and
- systems management at the boundary between the data-link layer and systems management of the fieldbus reference model.

1.2 Specifications

The principal objective of this standard is to specify the characteristics of conceptual data-link layer services suitable for time-critical communications, and thus supplement the OSI Basic Reference Model in guiding the development of data-link protocols for time-critical communications. A secondary objective is to provide migration paths from previously-existing industrial communications protocols.

This specification may be used as the basis for formal DL-Programming-Interfaces. Nevertheless, it is not a formal programming interface, and any such interface will need to address implementation issues not covered by this specification, including

- a) the sizes and octet ordering of various multi-octet service parameters, and
- b) the correlation of paired request and confirm, or indication and response, primitives.

1.3 Conformance

This standard do not specify individual implementations or products, nor do they constrain the implementations of data-link entities within industrial automation systems.

There is no conformance of equipment to this data-link layer service definition standard. Instead, conformance is achieved through implementation of the corresponding data-link protocol that fulfills the Type 11 data-link layer services defined in this standard.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61158-4-11, *Industrial communication networks – Fieldbus specifications – Part 4-11: Data-link layer protocol specification – Type 11 elements*

ISO/IEC 7498-1, *Information technology – Open Systems Interconnection – Basic Reference Model: The Basic Model*

ISO/IEC 7498-3, *Information technology – Open Systems Interconnection – Basic Reference Model: Naming and addressing*

ISO/IEC 8802-3, *Information technology – Telecommunications and information exchange between systems – Local and metropolitan area networks – Specific requirements – Part 3: Carrier sense multiple access with collision detection (CSMA/CD) access method and physical layer specifications*

ISO/IEC 10731, *Information technology – Open Systems Interconnection – Basic Reference Model – Conventions for the definition of OSI services*

ISO/TR 13283, *Industrial automation – Time-critical communications architectures – User requirements and network management for time-critical communications systems*

3 Terms, definitions, symbols, abbreviations and conventions

For the purposes of this document, the following terms, definitions, symbols, abbreviations and conventions apply.

3.1 Reference model terms and definitions

This standard is based in part on the concepts developed in ISO/IEC 7498-1 and ISO/IEC 7498-3, and makes use of the following terms defined therein.

3.1.1 DL-address	[7498-3]
3.1.2 DL-address-mapping	[7498-1]
3.1.3 called-DL-address	[7498-3]
3.1.4 calling-DL-address	[7498-3]
3.1.5 centralized multi-end-point-connection	[7498-1]
3.1.6 DL-connection	[7498-1]
3.1.7 DL-connection-end-point	[7498-1]
3.1.8 DL-connection-end-point-identifier	[7498-1]
3.1.9 DL-connection-mode transmission	[7498-1]
3.1.10 DL-connectionless-mode transmission	[7498-1]
3.1.11 correspondent (N)-entities	[7498-1]
correspondent DL-entities (N=2)	
correspondent Ph-entities (N=1)	

3.1.12 DL-duplex-transmission	[7498-1]
3.1.13 (N)-entity	[7498-1]
DL-entity (N=2)	
Ph-entity (N=1)	
3.1.14 DL-facility	[7498-1]
3.1.15 flow control	[7498-1]
3.1.16 (N)-layer	[7498-1]
DL-layer (N=2)	
Ph-layer (N=1)	
3.1.17 layer-management	[7498-1]
3.1.18 DL-local-view	[7498-3]
3.1.19 DL-name	[7498-3]
3.1.20 naming-(addressing)-domain	[7498-3]
3.1.21 peer-entities	[7498-1]
3.1.22 primitive name	[7498-3]
3.1.23 DL-protocol	[7498-1]
3.1.24 DL-protocol-connection-identifier	[7498-1]
3.1.25 DL-protocol-data-unit	[7498-1]
3.1.26 DL-relay	[7498-1]
3.1.27 reset	[7498-1]
3.1.28 responding-DL-address	[7498-3]
3.1.29 routing	[7498-1]
3.1.30 segmenting	[7498-1]
3.1.31 (N)-service	[7498-1]
DL-service (N=2)	
Ph-service (N=1)	
3.1.32 (N)-service-access-point	[7498-1]
DL-service-access-point (N=2)	
Ph-service-access-point (N=1)	
3.1.33 DL-service-access-point-address	[7498-3]
3.1.34 DL-service-connection-identifier	[7498-1]
3.1.35 DL-service-data-unit	[7498-1]
3.1.36 DL-simplex-transmission	[7498-1]
3.1.37 DL-subsystem	[7498-1]
3.1.38 systems-management	[7498-1]
3.1.39 DLS-user-data	[7498-1]

3.2 Service convention terms and definitions

This standard also makes use of the following terms defined in ISO/IEC 10731 as they apply to the data-link layer:

- 3.2.1 acceptor**
- 3.2.2 asymmetrical service**
- 3.2.3 confirm (primitive);
requestor.deliver (primitive)**
- 3.2.4 deliver (primitive)**
- 3.2.5 DL-confirmed-facility**
- 3.2.6 DL-facility**
- 3.2.7 DL-local-view**
- 3.2.8 DL-mandatory-facility**
- 3.2.9 DL-non-confirmed-facility**
- 3.2.10 DL-provider-initiated-facility**
- 3.2.11 DL-provider-optional-facility**
- 3.2.12 DL-service-primitive;
primitive**
- 3.2.13 DL-service-provider**
- 3.2.14 DL-service-user**
- 3.2.15 DLS-user-optional-facility**
- 3.2.16 indication (primitive);
acceptor.deliver (primitive)**
- 3.2.17 multi-peer**
- 3.2.18 request (primitive);
requestor.submit (primitive)**
- 3.2.19 requestor**
- 3.2.20 response (primitive);
acceptor.submit (primitive)**
- 3.2.21 submit (primitive)**
- 3.2.22 symmetrical service**

3.3 Data-link service terms and definitions

3.3.1

common memory

virtual common memory over the Type 11 fieldbus, which is shared by the nodes participating in the Type 11 fieldbus and is primarily used for the real-time communications by the TCC data service

3.3.2

DL-segment, link, local link

single DL-subnetwork in which any of the connected DLEs may communicate directly, without any intervening DL-relaying, whenever all of those DLEs that are participating in an instance of communication are simultaneously attentive to the DL-subnetwork during the period(s) of attempted communication

3.3.3

DLCEP-address

DL-address which designates either

a) one peer DL-connection-end-point, or

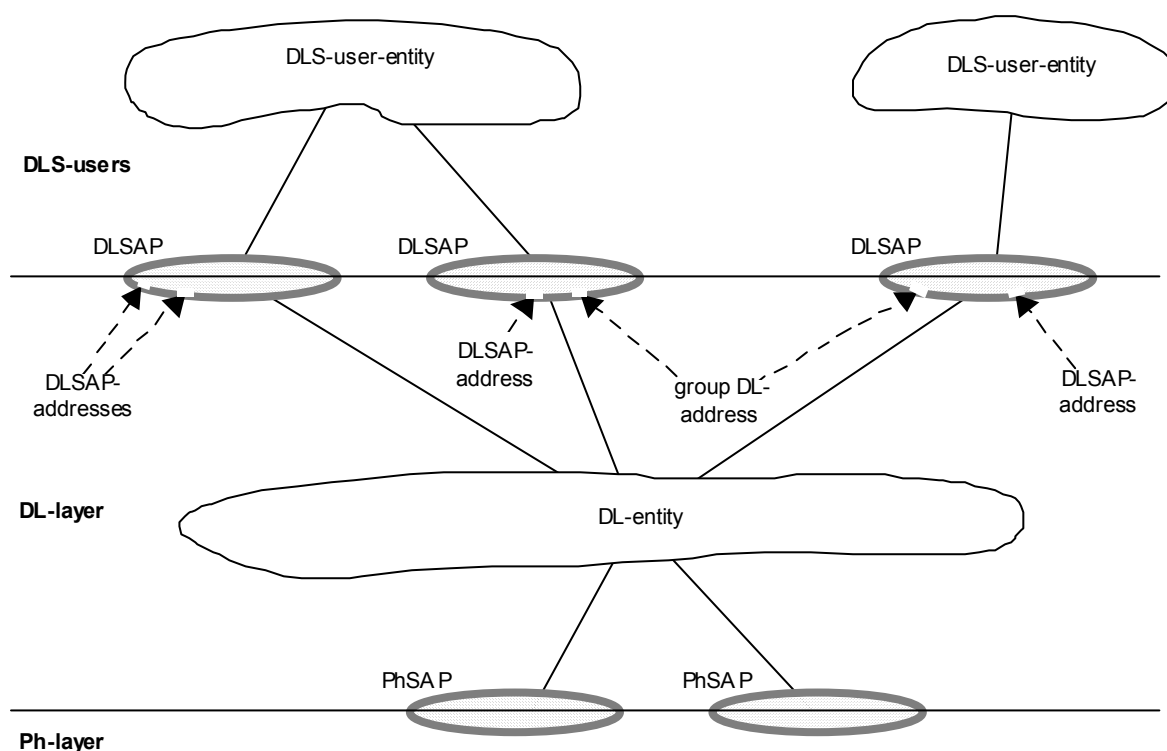
b) one multi-peer publisher DL-connection-end-point and implicitly the corresponding set of subscriber DL-connection-end-points where each DL-connection-end-point exists within a distinct DLSAP and is associated with a corresponding distinct DLSAP-address

3.3.4

DLSAP

distinctive point at which DL-services are provided by a single DL-entity to a single higher-layer entity

NOTE This definition, derived from ISO/IEC 7498-1, is repeated here to facilitate understanding of the critical distinction between DLSAPs and their DL-addresses.



NOTE 1 DLSAPs and PhSAPs are depicted as ovals spanning the boundary between two adjacent layers.

NOTE 2 DL-addresses are depicted as designating small gaps (points of access) in the DLL portion of a DLSAP.

NOTE 3 A single DL-entity may have multiple DLSAP-addresses and group DL-addresses associated with a single DLSAP.

Figure 1 – Relationships of DLSAPs, DLSAP-addresses and group DL-addresses

3.3.5

DL(SAP)-address

either an individual DLSAP-address, designating a single DLSAP of a single DLS-user, or a group DL-address potentially designating multiple DLSAPs, each of a single DLS-user

NOTE This terminology is chosen because ISO/IEC 7498-3 does not permit the use of the term DLSAP-address to designate more than a single DLSAP at a single DLS-user.

3.3.6

(individual) DLSAP-address

DL-address that designates only one DLSAP within the extended link

NOTE A single DL-entity may have multiple DLSAP-addresses associated with a single DLSAP.

3.3.7

extended link

DL-subnetwork, consisting of the maximal set of links interconnected by DL-relays, sharing a single DL-name (DL-address) space, in which any of the connected DL-entities may communicate, one with another, either directly or with the assistance of one or more of those intervening DL-relay entities

NOTE An extended link may be composed of just a single link.

3.3.8

frame

denigrated synonym for DLPDU

3.3.9

group DL-address

DL-address that potentially designates more than one DLSAP within the extended link. A single DL-entity may have multiple group DL-addresses associated with a single DLSAP. A single DL-entity also may have a single group DL-address associated with more than one DLSAP

3.3.10

high-speed cyclic data

RTE data conveyed by means of the high-speed cyclic data transmission

3.3.11

high-speed cyclic data transmission

one of three levels of the TCC data service with the highest priority level

3.3.12

low-speed cyclic data

RTE data conveyed by means of the low-speed cyclic data transmission

3.3.13

low-speed cyclic data transmission

one of three levels of the TCC data service with the lowest priority level

3.3.14

medium-speed cyclic data

RTE data conveyed by means of the medium-speed cyclic data transmission

3.3.15

medium-speed cyclic data transmission

one of three levels of the TCC data service with the second priority level

3.3.16

multipoint connection

connection from one node to many nodes. Multipoint connection allows data transfer from a single publisher to many subscriber nodes.

3.3.17

multi-peer DLC

centralized multi-end-point DL-connection offering DL-duplex-transmission between a single distinguished DLS-user known as the publisher or publishing DLS-user, and a set of peer but undistinguished DLS-users known collectively as the subscribers or subscribing DLS-users, where the publishing DLS-user can send to the subscribing DLS-users as a group (but not

individually), and the subscribing DLS-users can send to the publishing DLS-user (but not to each other).

3.3.18

node

single DL-entity as it appears on one local link

3.3.19

node-id

two-octet primary identifier for the DLE on the local link, whose values are constrained

NOTE A permissible value is from 1 to 255. A value 0 is specifically used for the SYN node, which emits the SYN frame.

3.3.20

receiving DLS-user

DL-service user that acts as a recipient of DLS-user-data

NOTE A DL-service user can be concurrently both a sending and receiving DLS-user.

3.3.21

sending DLS-user

DL-service user that acts as a source of DLS-user-data

3.3.22

SYN node

node transmitting the SYN frame

3.3.23

sporadic message data service

aperiodic message transfer which sporadically occurs upon DLS-user requesting one or more message to transfer, and regular ISO/IEC 8802-3 Ethernet message frame is transferred by means of this message transfer

3.3.24

TCC data service

cyclic data transfer with three levels of the data transmission at the same time, of which each data transmission level is according to the data priority and the data transmission period for real-time delivery, and of which the data transmission period and the total data volume for each level can be specified in designing phase and on application needs

3.4 Symbols and abbreviations

3.4.1 CM	Common memory
3.4.2 DL-	Data-link layer (as a prefix)
3.4.3 DLC	DL-connection
3.4.4 DLCEP	DL-connection-end-point
3.4.5 DLE	DL-entity (the local active instance of the data-link layer)
3.4.6 DLL	DL-layer
3.4.7 DLPCI	DL-protocol-control-information
3.4.8 DLPDU	DL-protocol-data-unit
3.4.9 DLM	DL-management

3.4.10 DLME	DL-management entity (the local active instance of DL-management)
3.4.11 DLMS	DL-management service
3.4.12 DLS	DL-service
3.4.13 DLSAP	DL-service-access-point
3.4.14 DLSDU	DL-service-data-unit
3.4.15 FIFO	First-in first-out (queuing method)
3.4.16 GCM	Global common memory
3.4.17 OSI	Open systems interconnection
3.4.18 Ph-	Physical layer (as a prefix)
3.4.19 PhE	Ph-entity (the local active instance of the physical layer)
3.4.20 PhL	Ph-layer
3.4.21 QoS	Quality of service
3.4.22 RTE	Real Time Ethernet
3.4.23 TCC	Time-critical cyclic

3.5 Common conventions

This standard uses the descriptive conventions given in ISO/IEC 10731.

The service model, service primitives, and time-sequence diagrams used are entirely abstract descriptions; they do not represent a specification for implementation.

Service primitives, used to represent service user/service provider interactions (see ISO/IEC 10731), convey parameters that indicate information available in the user/provider interaction.

This standard uses a tabular format to describe the component parameters of the DLS primitives. The parameters that apply to each group of DLS primitives are set out in tables throughout the remainder of this standard. Each table consists of up to six columns, containing the name of the service parameter, and a column each for those primitives and parameter-transfer directions used by the DLS:

- a) the request primitive's input parameters;
- b) the request primitive's output parameters;
- c) the indication primitive's output parameters;
- d) the response primitive's input parameters; and
- e) the confirm primitive's output parameters.

NOTE The request, indication, response and confirm primitives are also known as requestor.submit, acceptor.deliver, acceptor.submit, and requestor.deliver primitives, respectively (see ISO/IEC 10731).

One parameter (or part of it) is listed in each row of each table. Under the appropriate service primitive columns, a code is used to specify the type of usage of the parameter on the primitive and parameter direction specified in the column:

M Parameter: mandatory for the primitive.

- U** Parameter: a user option which may or may not be provided depending on the dynamic usage of the DLS-user. When not provided, a default value for the parameter is assumed.
- C** Parameter is conditional upon other parameters or upon the environment of the DLS-user.
- (Blank) Parameter is never present.

Some entries are further qualified by items in brackets. These may be

- a) a parameter-specific constraint
- (=) indicates that the parameter is semantically equivalent to the parameter in the service primitive to its immediate left in the table.
- b) an indication that some note applies to the entry
- (n) indicates that the following note n contains additional information pertaining to the parameter and its use.

In any particular interface, not all parameters need be explicitly stated. Some may be implicitly associated with the DLSAP at which the primitive is issued.

In the diagrams which illustrate these interfaces, dashed lines indicate cause-and-effect or time-sequence relationships, and wavy lines indicate that events are roughly contemporaneous.

4 Data-link service and concept

4.1 Overview

4.1.1 General

This standard specifies the Type 11 data-link services for a ISO/IEC 8802-3-based time-critical control network, which is one of the communication networks for Real-Time Ethernet (RTE) defined in IEC 61784-2.

This standard meets the industrial automation market objective of providing predictable time deterministic and reliable time-critical data transfer and means, which allow co-existence with non-time-critical data transfer over the ISO/IEC 8802-3 series communications medium, for support of cooperation and synchronization between automation processes on field devices in a real-time application system. The term “time-critical” is used to represent the presence of a time-window, within which one or more specified actions are required to be completed with some defined level of certainty.

4.1.2 Field of applications

In industrial control systems, several kinds of field devices such as drives, sensors and actuators, programmable controllers, distributed control systems (DCS) and human machine interface (HMI) devices are required to be connected with control networks. The process control data and the state data is transferred among these field devices in the system and the communications between these field devices requires simplicity in application programming and to be executed with adequate response time. In most industrial automation systems such as food, water, sewage, paper and steel, including a rolling mill, the control network is required to provide time-critical response capability for their application, as required in ISO/TR 13283 for time-critical communications architectures.

Plant production may be compromised due to errors, which could be introduced to the control system if the network does not provide a time-critical response. Therefore, the following characteristics are required for a time-critical control network:

- deterministic response time between the control device nodes;
- ability to share process data seamlessly across the control system.

These services are applicable to such an industrial automation environment, in which time-critical communications is primarily required. The term “time-critical” is used to represent the presence of a time window, within which one or more specified actions are required to be completed with some defined level of certainty. Failure to complete specified actions within the time-window risks failure of the applications requesting the actions, with attendant risk to equipment, plant and possibly human life.

4.2 General description of services

4.2.1 General

The DLS provides for transparent and reliable transfer of data between DLS-users over a Type 11 fieldbus. The DLS is based on services provided by the physical layer of ISO/IEC 8802-3 to the conceptual interface between the physical and data-link layers.

Two types of data transmission services are provided.

- a) **TCC data service:** The connection-oriented buffer transfers between pre-established point-to-multipoint DLCEPs on the same local link.
- b) **Sporadic message data service:** The unacknowledged connectionless message transfers between single DLSAPs, or unacknowledged connectionless message transfer from a single DLSAP to a group of DLSAPs on the extended link.

NOTE For the purpose of clarity, the expressions “buffer transfer” and “message transfer” are used to distinguish between the two types of communications services, connection-oriented and connectionless, respectively, that are offered by this DLS.

The buffer transfer service or the TCC data service is based on cyclic data transfers of three kinds of transmission period. The transmission periods and the total volumes of each level of the buffer transfer by the multiple distributed DLS-providers on the same local link are defined when the system is configured, and are based on application needs. Cyclic data transfers are automatically triggered by the communications system without the user requesting them.

There are also three types of TCC data service according to the transmission period.

- a) High-speed cyclic data transmission.
- b) Medium-speed cyclic data transmission.
- c) Low-speed cyclic data transmission.

The message transfer service or the sporadic message data service is based on aperiodic data transfer, sporadically occurred upon DLS-user requesting one or more message to transfer. The priority level and the target token rotation time, corresponding to the target time to obtain the transmission right to send out the message data on the medium, are defined when the system is configured, and are based on application needs. The transmit delay is depending on the priority level and the target token rotation time. Regular ISO/IEC 8802-3 Ethernet message frame is transferred by means of this message transfer service.

The DLS provides DLS-users with a means to set up a quality of service for the transfer of data. QoS is specified by means of QoS parameters representing aspects such as data transmit delay, priority level, DLCEP data length, and so on. A DL-management Service (DLMS) is defined in Clause 5.

4.2.2 Overview of the Data frames flow on the medium

Overview of the data frames flow on the medium is shown in Figure 2 and Figure 3.

The DLL provides the opportunity of transferring data to each node in a sequential order and within a predetermined time period. At the start time of every high-speed-transmission-period (T_{syn}), the SYN frame is broadcast to all Type 11 fieldbus nodes. When the SYN frame has been received, the node with sequential number 1 (node 1) starts sending its data frames, and after the completion of its data frames transmission node 2 can send out its data frames. The Nth node (node N) can obtain the transmission right after the (N-1) th node completes its data frames transmission. The sequential number is assigned to each node at the time approval to join the Type 11 fieldbus is granted, and is up to 255.

Each node can hold the transmission right for a preset time before transferring the transmission right to the next node. The data to be sent and the data to be retained are determined by priority.

Data transmission includes cyclic data and sporadic message transmission. Cyclic data transmission is divided into high, medium and low-speed cyclic data transmission. Each node sends the high-speed cyclic data frames on each occasion when it obtains the transmission right. The data of lower priorities, that is the medium-speed cyclic data, the sporadic message data and the low-speed cyclic data respectively, are sent or not sent depending on the circumstances.

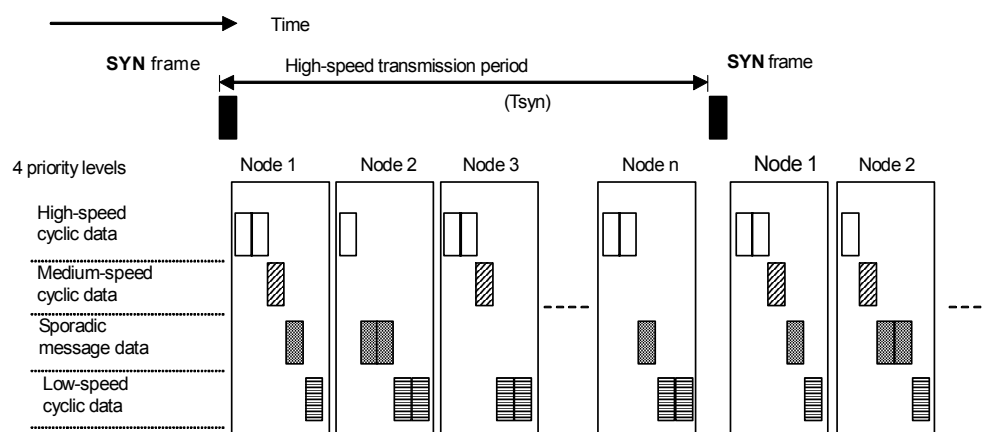


Figure 2 – Overall flow of data frames during one minimum cycle period (high-speed transmission period)

The holding time of the transmission right of each node is determined by the settings of the high-speed cyclic, the medium-speed cyclic, the sporadic message and the low-speed cyclic data transmission periods and by the volume of transmission data for each node. After sending all the high-speed cyclic data, the node sends the medium-speed cyclic data. If the holding time of the transmission right ends during sending the medium-speed cyclic data, the transmission of the medium-speed cyclic data is interrupted. Node N obtains the transmission right again during the next high-speed transmission period, during which time all the high-speed cyclic data and the remainder of the previous medium-speed cyclic data is sent. For the low-speed cyclic data are sent out in this fashion.

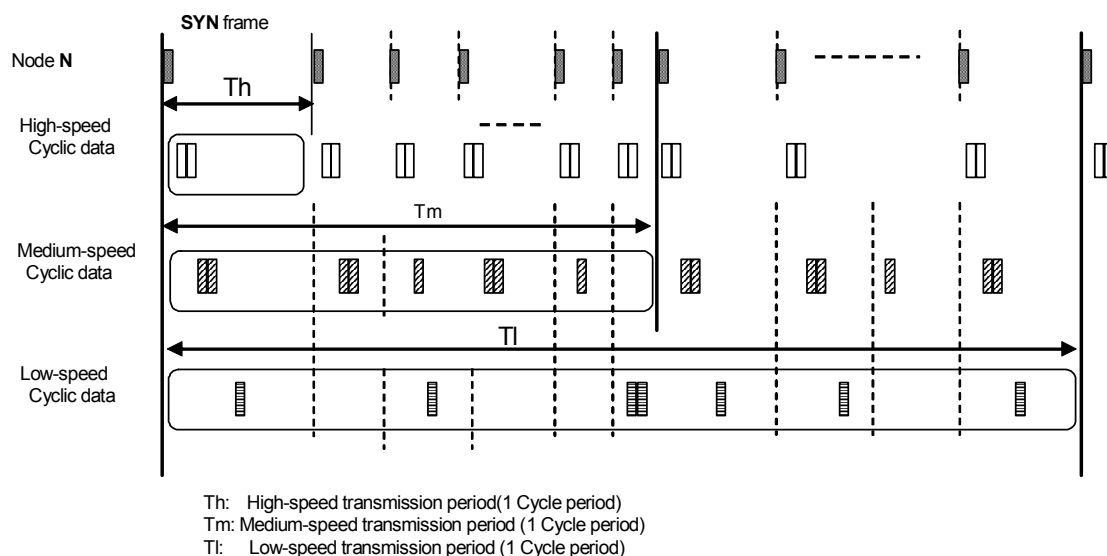


Figure 3 – Overall flow of cyclic data frames over one maximum cycle period (low-speed transmission period)

4.2.3 Sequence of primitives

A request primitive is used by a DLS-user to request a service. A confirm primitive is returned to the DLS-user upon completion of the service. Moreover, an indication primitive is used to report to the DLS-user the receipt of new DLS-user data or the receipt of a new message.

4.2.3.1 Primitives of the TCC data service

The sequence of primitives for the TCC data service is shown in Figure 4.

DL-Data-req indication primitive informs the DLS-user initiates the data transfer using DL-Put service from the corresponding DLS-user buffer associated with a specified DLCEP to the send buffer of local DLE for publishing the data.

DL-Put request primitive which responds to DL-Data-req indication primitive allows the DLS-user to transfer data of the corresponding DLS-user buffer associated with a specified DLCEP by the DL-Data-req indication primitive to the send_buffer of the local DLE where the DLE is the publisher.

DL-Buffer-received indication primitive informs the DLS-user that a subscribed data if a DLCEP has just correctly received. The data in the Receive_buffer of the local DLE is updated and is available, and is read by the DLS-user using the DL-Get request primitive.

DL-Get request primitive which responds to DL-Buffer-received indication primitive allows the DLS-user to get the data in the receive_buffer of the local DLE and to transfer the data from the receive_buffer to the corresponding DLS-user buffer associated with the specified DLCEP by DL-Buffer-received indication primitive by the DLE where the DLE is the subscriber.

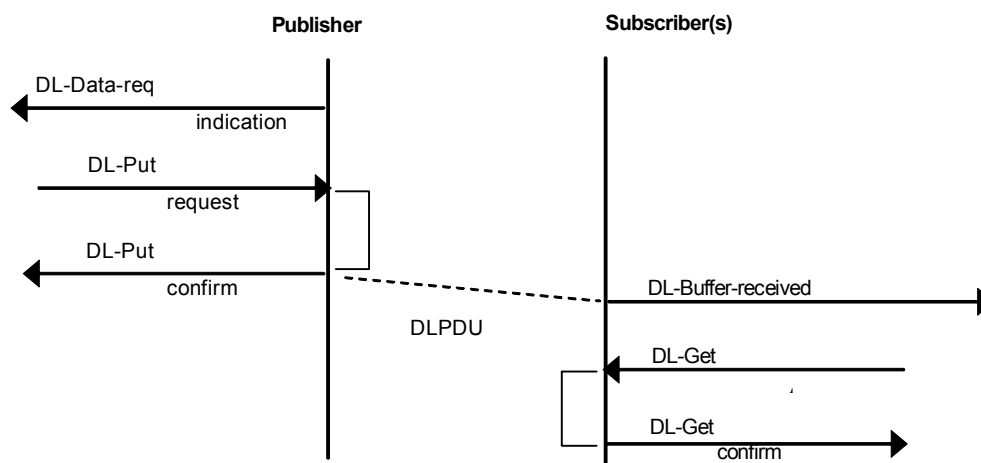


Figure 4 – Sequence diagram of TCC data service

4.2.3.2 Sporadic message data service

The sequence of primitives on sporadic message data service is shown in Figure 5.

DL-SPDATA request and DL-SPDATA indication correspond to the MA_DATA request and MA_DATA indication defined by ISO/IEC 8802-3 respectively.

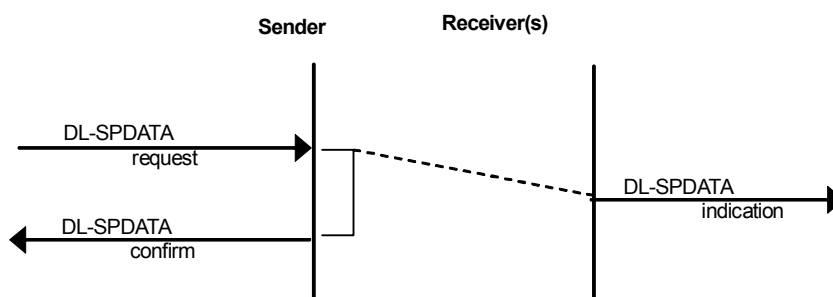


Figure 5 – Sequence diagram of sporadic message service

4.2.4 Addressing

Two different types of addressing exist for DL-addressing shown in Figure 6: one for the TCC data service and the other for the sporadic message data service.

For buffer transfers of the TCC data service: each variable in the system is associated with a DLCEP-identifier that characterizes it within the system in a unique manner. Entities participating in a buffer transfer are not identified explicitly. Rather, they are identified indirectly as subscribers or publisher of the identified variable. Each variable has only one publisher.

For message transfer of the sporadic message data service: One or more DLSAP-addresses are defined within each DLE. These DLSAP-addresses give access to a message transfer service. Each DLSAP-address identifies an access point to a message service linked to a DLS-user entity. During the message transaction two DLSAP-addresses are indicated in order to establish contact between the communicating entities. Each DLSAP-address specifies a DLS-user of the message service (for both transmission and reception). This DL-address is unique within the extended link.

Variable addressing is restricted to the local link. The addressing mechanism makes it possible to identify variables and exchanges independent of the producing and consuming DLEs. For buffer transfers all relationships between the various DLS-users are known and defined when the system is configured. Each DLCEP-identifier characterizes a single system variable and thus establishes a relationship between the unique publisher of the variable and the subscribers of the variable.

Buffer transfers use the local broadcast medium and are restricted to the local link: the DLCEP-identifier and the value of a variable are made available to all DLEs on the local link. The DLCEP-identifier associated with the variable allows each DLE to recognize whether or not it is the publisher or a subscriber of the value associated with the identified variable.

The relationship of peer to peer and multi-peer DLCs and their DLCEPs is shown in Figure 6. DLSAPs and PhSAPs are depicted as ovals spanning the boundary between two adjacent layers. DL-addresses are depicted as designating small gaps (points of access) in the DLL portion of a DLSAP. A DLCEP-address also designates a specific point of information flow (its DLCEP) within the DLSAP. A single DL-entity may have multiple DLSAP-addresses and group DL-addresses associated with a single DLSAP. This figure also shows the relationships of DL-paths and PhSAPs.

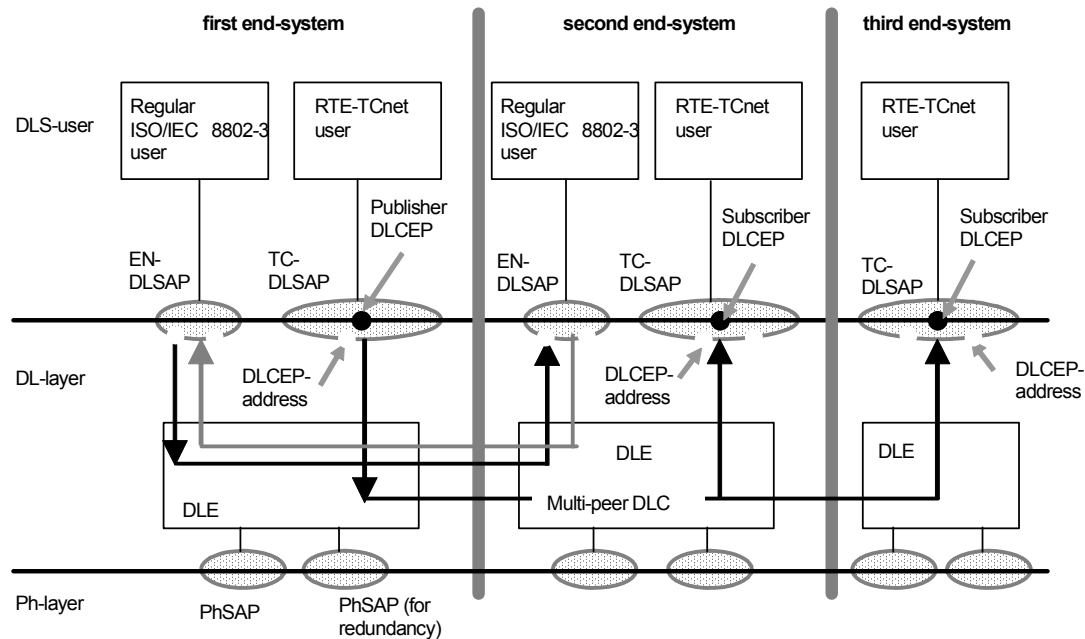


Figure 6 – Relationship of DLSAP, DLCEP and DLCEP-address

4.3 TCC data service

4.3.1 General

This DLS provides connection-oriented buffer transfers between pre-established point-to-multipoint DLCEPs on the same local link.

4.3.2 Sequence of primitives

The TCC Data service primitives and the parameters are summarized in Table 1, and the primitive sequence is shown in Figure 4.

Table 1 – Primitives and parameters used on the time-critical cyclic data service

Function	Location	Primitive	Direction	Parameters
Notify publisher DLS-user	Publisher	DL-Data-req indication	From DLE	DLCEP-identifier DLSDU-length
Put buffer	Publisher	DL-Put request	To DLE	DLCEP-identifier DLSDU-length DLSDU
Put confirm to publisher DLS-user	Publisher	DL-Put confirm	From DLE	DLCEP-identifier Status
Notify subscriber DLS-user	Subscriber(s)	DL-Buffer-received indication	From DLE	DLCEP-identifier DLSDU-length
Get buffer	Subscriber(s)	DL-Get request	To DLE	DLCEP-identifier DLSDU-length
Confirm to publisher DLS-user	Subscriber(s)	DL-Get confirm	From DLE	DLCEP-identifier DLSDU-length DLSDU Status
NOTE In this table, time increases from top to bottom.				

4.3.3 Data request

4.3.3.1 Function

DL-Data-req service informs the DLS-user that initiates the transfer using the DL-Put service from the corresponding DLS-user buffer associated with a specified DLCEP to the send_buffer of local DLE for publishing the data.

4.3.3.2 Types of primitives and the parameters of Data request

Table 2 indicates the parameters of Data request service.

Table 2 – Data request primitives and the parameters

DL-DATA_REQ	Indication
Parameter name	output
DLCEP-identifier	M
DLSDU-length	M

4.3.3.2.1 DLCEP-identifier

This parameter specifies from which buffer the DLS-user reads and transfers the data into the send_buffer of the local DLE for publishing.

4.3.3.2.2 DLSDU-length

This parameter indicates the length of the identified buffer, that is the actual length of the contents in a DLSDU.

4.3.4 Put buffer

4.3.4.1 Function

DL-Put request, which responds to DL-Data-req indication, allows the DLS-user to transfer data of the corresponding DLS-user buffer associated with the specified DLCEP by the DL-Data-req indication to the Send_buffer of the local DLE where the DLE is the publisher.

4.3.4.2 Types of primitives and parameters of Put buffer

Table 3 indicates the types of primitives and parameters for Put buffer.

Table 3 – Put buffer primitives and parameters

DL-Put	Request	Confirm
Parameter name	input	output
DLCEP-identifier	M	M(=)
DLSDU-length	M	M(=)
DLSDU	M	
Status		M

4.3.4.2.1 DLCEP-identifier

This parameter indicates from which buffer DLS-user reads and writes data into the send_buffer of the local DLE.

4.3.4.2.2 DLSDU-length

This parameter indicates the length of the identified buffer, that is the actual length of the contents in a DLSDU.

4.3.4.2.3 DLSDU

This parameter specifies the information that is transferred by buffer transfer from the local DLE as a publisher to the remote multi-peer DLEs as subscribers.

4.3.4.2.4 Status

This parameter allows the DLS-user to determine whether the requested DLS was provided successfully, or failed for the reason specified. The value conveyed in this parameter is as follows:

- a) "success";
- b) "failure – invalid requested parameter".

4.3.5 Get buffer

4.3.5.1 Function

DL-Get request which responds to DL-Buffer-received indication allows the DLS-user to get and transfer the data from the receive_buffer of the local DLE to the corresponding DLS-user buffer associated with the specified DLCEP by DL-Buffer-received indication by the DLE where the DLE is the subscriber.

4.3.5.2 Type of primitives and parameters of the Get buffer

Table 4 indicates the types of primitives and parameters for Get buffer.

Table 4 – Get buffer primitives and parameters

DL-Get	Request	Confirm
Parameter name	input	output
DLCEP-identifier	M	M(=)
DLSDU-length	M	M(=)
DLSDU		M
Status		M

4.3.5.2.1 DLCEP-identifier

This parameter unambiguously designates to which buffer DLS-user reads and transfers data from the receive_buffer of the local DLE.

4.3.5.2.2 DLSDU-length

This parameter indicates the length of the data to be transferred from the Receive_buffer of the local DLE to the identified buffer of the DLS-user.

4.3.5.2.3 DLSDU

This parameter specifies the information that is transferred by buffer transfer by the remote DLE as a publisher to the local DLE as subscribers.

4.3.5.2.4 Status

This parameter allows the DLS-user to determine whether the requested DLS was provided successfully, or failed for the reason specified. The value conveyed in this parameter is as follows:

- a) "success";
- b) "failure -- invalid requested parameter".

4.3.6 Notify buffer received**4.3.6.1 Function**

DL-Buffer-received indication informs the DLS-user that a subscribed identified data has just correctly received. The data in the receive_buffer of the local DLE is updated and is available, and is read by the DLS-user using the DL-Get request primitive.

4.3.6.2 Types of primitives and parameters of the Notify buffer received

Table 5 indicates the types of primitives and parameters for Notify buffer received.

Table 5 – Notify buffer received primitives and parameters

DL-Buffer_Received	Indication
Parameter name	output
DLCEP-identifier	M
DLSDU-length	M

4.3.6.2.1 DLCEP-identifier

This parameter unambiguously designates to which buffer DLS-user reads and transfers the data from the receive_buffer of the local DLE.

4.3.6.2.2 DLSDU-length

This parameter indicates the length of the identified buffer that is the actual length of the contents in a DLSDU.

4.4 Detail description of the sporadic message data service

4.4.1 General

This DLS provides unacknowledged connectionless message transfers between single DLSAPs, or unacknowledged connectionless message transfer from a single DLSAP to a group of DLSAPs on the extended link.

A pair set of DL-SPDATA request and DL-SPDATA indication corresponds to that of the MA_DATA request and MA_DATA indication defined by ISO/IEC 8802-3 respectively.

4.4.2 Sequence of primitives

The sporadic message data service primitives and the parameters are summarized in Table 6, and the primitive sequence is shown in Figure 5.

Table 6 – Primitives and parameters used on sporadic message data service

Function	Location	Primitive	Direction	Parameters
Send DLSDU	Sender	DL-SPDATA request	To DLE	DA(destination_address) SA(source_address) MSDU(m_sdu) Service-class
Send confirmation to calling DLS-user	Sender	DL-SPDATA confirm	From DLE	Status
Notify and provide MSDU received to peer called DLS-user	Receiver(s)	DL-SPDATA indication	From DLE	DA(destination_address) SA(source_address) MSDU(m_sdu) Rec-status(reception-status)
NOTE 1 In this table, time increases from top to bottom.				
NOTE 2 DL-SPDATA service primitives respond to the MA_DATA service primitives defined by ISO/IEC 8802-3.				

4.4.3 Send sporadic message

4.4.3.1 Function

DL-SPDATA service primitives allow the DLS-user to transfer message data to a single peer DLS-user or multi-peer DLS-users in remote nodes.

4.4.3.2 Types of primitives and parameters of the send sporadic message

Table 7 indicates the types of primitives and parameters for Send sporadic message.

Table 7 – Submit sporadic message primitives and parameters

DL-SPDATA	Request	Indication	Confirm
Parameter name	input	output	output
DA(destination address)	M	M(=)	M(=)
SA(source address)	U	M(=)	
Service-class	U		
MSDU (m_sdu)	M	M(=)	
Rec-status (reception status)		M	
Status			M

4.4.3.2.1 DA (destination address)

This parameter indicates the DLE(s) for which the DLPDU is intended. It may be an individual or multicast (including broadcast) address.

4.4.3.2.2 SA (source address)

This parameter indicates an individual DLE from which the DLPDU is sending.

4.4.3.2.3 MSDU (MAC service data unit)

This parameter specifies the MAC service data unit to be transmitted by MAC sublayer entity.

4.4.3.2.4 Service-class

This parameter indicates a quality of service requested by DLS-user.

NOTE The Type 11 DL protocol as well as the CSMA/CD (ISO/IEC 8802-3) MAC protocol provides a single quality of service regardless of the service-class.

4.4.3.2.5 Rec-status

This parameter is used to pass status information upon the DLPDU received to the DLS-user:

- a) "success – received without error";
- b) "failure – received but with error".

4.4.3.2.6 Status

This parameter allows the DLS-user to determine whether the requested DLS was provided successfully, or failed for the reason specified. The value conveyed in this parameter is as follows:

- a) "success – successfully completed";
- b) "failure – invalid requested parameter".

5 DL-management services

5.1 General

The interface between a DLE and a DL-management user (DLMS-user) is described. The services of this interface are needed for the protocol which implements the DLS specified in 3.5.

5.2 Facilities of the DL-management service

The DL-management service provides the facilities for the initialization, configuration, event and error handling between the DLMS-user and the logical functions in the DLE. The following functions are provided to the DLMS-user.

- a) Reset of the local DLE.
- b) Request for and modification of the actual operating parameters and of the counters of the local DLE.
- c) Notification of unexpected events, errors, and status changes, both local and remote.
- d) Request for identification and for the common memory configuration of the local DLE.
- e) Activation and deactivation of the cyclic data transmission of the local DLE.

5.3 Service of the DL-management

5.3.1 Overview

DL-management provides the following services to DLMS-user.

- a) Reset
- b) Set-Value
- c) Get-Value
- d) Event
- e) Set-Publisher-configuration
- f) Get-Publisher-configuration
- g) Activate-Publisher
- h) Deactivate-Publisher.

The services Reset, Set-Value, Event, Set-Publisher configuration and Activate-Publisher are considered mandatory. Other services are considered optional.

5.3.2 Reset

The DLMS-user employs this service to cause DL-management to reset the DLE. A reset is equivalent to power on. The DLMS-user receives a confirmation thereof.

5.3.3 Set value

The DLMS-user employs this service to assign new values to the variables of the DLE. The DLMS-user receives confirmation that the specified variables have been set to the new values.

5.3.4 Get value

This service allows the DLMS-user to read variables of the DLE. The response of the DL-management returns the actual value of the specified variables.

5.3.5 Event

DL-management employs this service to inform the DLMS-user about certain events or error in the DLL.

5.3.6 Set publisher configuration

The DLMS-user employs this service to assign new publisher configuration parameters to the TCC data of the DLE. DLS-user receives a confirmation whether the specified parameters have been set to the new one.

5.3.7 Get publisher configuration

This service enables DL-management to read publisher configuration parameters of the TCC data. The response of the DL-management returns the actual value of publisher configuration parameters.

5.3.8 Activate TCC data service

This service allows the DLMS-user to activate the TCC data service. The DLMS-user receives the confirmation whether the designated TCC data service has been started or this request has been completed unsuccessfully.

5.3.9 Deactivate TCC data service

This service allows the DLMS-user to deactivate the TCC data service. The DLMS-user receives the confirmation whether the designated TCC data service has been terminated or this request has been completed unsuccessfully.

5.4 Overview of interactions

The DL-management services and their primitives are summarized in Table 8.

Table 8 – Summary of DL-management primitives and parameters

Service	Primitive	Parameter
Reset	DLM-Reset request	(<none>)
	DLM-Reset confirm	(out Status)
Set value	DLM-Set-value request	(in Variable-name, Desired-value)
	DLM-Set-value confirm	(out Status)
Get value	DLM-Get-value request	(in Variable-name)
	DLM-Get-value confirm	(out Status, Current-value)
Event	DLM-Event indication	(out Event-identifier, Additional-information)
Set publisher configuration parameters	DLM-Set-publisher-configuration request	(in Desired-speed-class, Desire-configuration)
	DLM-Set-publisher-configuration confirm	(out Status)
Get publisher configuration parameters	DLM-Get-publisher-configuration request	(in Desire-speed-class)
	DLM-Get-publisher-configuration confirm	(out Status, Current-configuration)
Activate TCC service	DLM-Activate-TCC-data request	(in Desired-speed-class)
	DLM-Activate-TCC-data confirm	(out Status)
Deactivate TCC service	DLM-Deactivate-TCC-data request	(in Desired-speed-class)
	DLM-Deactivate-TCC-data confirm	(out Status)

The sequence of the DL-management primitives are shown in Figure 7 and Figure 8.

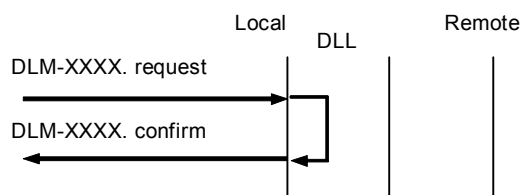


Figure 7 – Sequence diagram of Reset, Set-value, Get-value, Set-publisher-configuration, Get-publisher-configuration, Activate-TCC-data and Deactivate-TCC-data service primitives

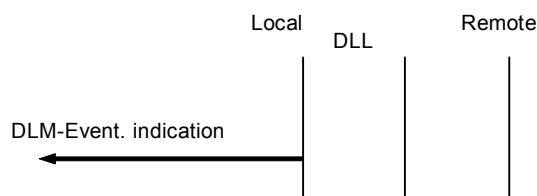


Figure 8 – Sequence diagram of sporadic message service

5.5 Detail specification of service and interactions

5.5.1 Reset

5.5.1.1 Function

The DLM-Reset request primitive causes DLMS to reset the DLE. The DLE assumes the “Offline” status after carried out in the same manner as at a power on, and all DLE variables are cleared. The DLMS-user receives the DLM-Reset confirmation primitive with the status of the result in success or failure.

5.5.1.2 Type of primitives and parameters of the DLM-Reset

Table 9 indicates the primitive and parameters of the Reset service.

Table 9 – DLM-Reset primitives and parameters

DLM-Reset	Request	Confirm
Parameter name	input	output
Status		M
NOTE The method by which a confirm primitive is correlated with its corresponding preceding request primitive is a local matter.		

5.5.1.2.1 Status

This parameter allows the DLMS-user to determine whether the requested DLMS was provided successfully, or failed for the reason specified. The value conveyed in this parameter is as follows:

- a) “OK – successfully completed”;
- b) “Failure – terminated before completion”.

5.5.2 Set-value

5.5.2.1 Function

This service is used to assign new values to the variables of the DLE. The DLMS-user receives confirmation that the specified variables has been set to the new values.

5.5.2.2 Type of primitives and parameters of DLM-Set-value

Table 10 indicates the primitive and parameters of the set service.

Table 10 – DLM-Set-value primitives and parameters

DLM-Set value	Request	Confirm
Parameter name	input	output
Variable-name	M	
Desired-value	M	
Status		M
NOTE The method by which a confirm primitive is correlated with its corresponding preceding request primitive is a local matter.		

5.5.2.2.1 Variable-name

This parameter specifies the variable within the DLE whose value is to be set.

NOTE The selectable variables and their permitted values or value ranges are defined in IEC 61158-4-11.

5.5.2.2.2 Desired-value

This parameter specifies the desired value for the selected variable.

5.5.2.2.3 Status

This parameter allows the DLMS-user to determine whether the requested DLMS was provided successfully, or failed for the reason specified. The value conveyed in this parameter is as follows:

- a) "OK – success – the variable could be update";
- b) "Failure – the variable does not exist or could not assume the new value";
- c) "Failure – invalid parameters in the request".

5.5.3 Get-value

5.5.3.1 Function

This service can be used to read the value of a DLE variable. The response of the DLMS returns the actual value of the specified variable.

5.5.3.2 Type of primitives and parameters of DLM-Get-value

Table 11 indicates the primitive and parameters of the DLM-Get-value service.

Table 11 – DLM-Get-value primitives and parameters

DLM-Get value	Request	Confirm
Parameter name	input	output
Variable-name	M	
Desired-value		M
Status		M
NOTE The method by which a confirm primitive is correlated with its corresponding preceding request primitive is a local matter.		

5.5.3.2.1 Variable-name

This parameter specifies the variable within the DLE whose value is being requested. The selectable variables are defined in the corresponding part of IEC 61158-4-11.

5.5.3.2.2 Status

This parameter allows the DLMS-user to determine whether the requested service was provided successfully, or failed for the reason specified. The value conveyed in this parameter is as follows:

- a) "OK – success – the variable could be read";
- b) "Failure – the variable does not exist or could not be read";
- c) "Failure – invalid parameters in the request".

5.5.3.2.3 Current-value

This parameter is present when the status parameter indicates that the requested service was performed successfully. This parameter specifies the current value of the selected variable.

NOTE The observable variables and their permitted value ranges are defined in IEC 61158-4-11.

5.5.4 Event

5.5.4.1 Function

This service is used to inform the DLMS-user about certain events or errors in the DLL.

5.5.4.2 Type of primitives and parameters of DLM-Event

Table 12 indicates the primitive and parameters of the DLM-Event service.

Table 12 – Event primitives and parameters

DLM-Event	Indication
Parameter name	output
DLM-event-identifier	M
Additional-information	C

5.5.4.2.1 DLM-event-identifier

This parameter specifies the primitive or composite event within the DLE whose occurrence is being announced.

NOTE The possible values are defined in the corresponding part of IEC 61158-4-11,

5.5.4.2.2 Additional-information

This optional parameter provides event-specific additional information.

5.5.5 Set-publisher-configuration

5.5.5.1 Function

This service allows the DLMS-user to set up the publisher configuration information related to the DLCEP in the DLL. The DLMS-user receives the confirmation when this service has been completed.

5.5.5.2 Type of primitives and parameters of the set-publisher-configuration

Table 13 indicates the primitive and parameters of the set-publisher-configuration service.

Table 13 – Set-publisher-configuration primitives and parameters

DLM-Set-publisher-configuration	Request	Confirm
Parameter name	input	output
Desired-speed-class	M	
Desired-configuration	M	
Status		M
NOTE The method by which a confirm primitive is correlated with its corresponding preceding request primitive is a local matter.		

5.5.5.2.1 Desired-speed-class

This parameter specifies the speed class for TCC data transmission. The DMLS-user can specify high-speed, medium-speed or low-speed for the speed class.

- a) High-speed
- b) Medium-speed
- c) Low-speed.

5.5.5.2.2 Desired-configuration

This parameter specifies a set of the DLCEP-identifiers as a publisher for the TCC data service with the desired-speed-class. The DLCEP-identifiers may be specified within the range of 0 to 2 047, though the maximum number can be specified in the limitation of $V(PB_h)$, $V(PB_m)$ or $V(PB_l)$ for the corresponding speed-class of high-speed, medium-speed or low-speed respectively.

5.5.5.2.3 Status

This parameter allows the DLMS-user to determine whether the requested service was provided successfully, or failed for the reason specified. The value conveyed in this parameter is as follows:

- a) "OK –success – the publisher configuration is updated";
- b) "Failure – terminated unsuccessfully";
- c) "Failure – invalid parameters in the request".

5.5.6 Get-publisher-configuration

5.5.6.1 Function

This option service allows the DLMS-user to read the publisher configuration information in the DLL. The DLMS-user receives the confirmation with the current publisher configuration when this service has been completed.

5.5.6.2 Type of primitives and parameters of get-publisher-configuration

Table 14 indicates the primitive and parameters of the get-publisher-configuration service.

Table 14 – DLM-Get-publisher-configuration primitives and parameters

DLM-Get-publisher-configuration	Request	Confirm
Parameter name	input	output
Desire-speed-class	M	
Status		M
Current-configuration		M
NOTE The method by which a confirm primitive is correlated with its corresponding preceding request primitive is a local matter.		

5.5.6.2.1 Desire-speed-class

This parameter specifies the speed class for the TCC data service. The DMLS-user can specify high-speed, medium-speed or low-speed for the speed class.

- a) High-speed
- b) Medium-speed
- c) Low-speed.

5.5.6.2.2 Status

This parameter allows the DLMS-user to determine whether the requested service was provided successfully, or failed for the reason specified. The value conveyed in this parameter is as follows:

- a) "Success — the current configuration could be read";
- b) "Failure — the current configuration does not exist or terminated unsuccessfully";
- c) "Failure — invalid parameters in the request".

5.5.6.2.3 Current-configuration

This parameter is present when the status parameter indicates that the requested service was performed successfully. This parameter indicates a set of the current values of the DLCEP-identifiers for the TCC data service of the corresponding desired-speed-class.

5.5.7 Activate TCC data service

5.5.7.1 Function

This service allows the DLMS-user to activate the TCC data service. The DLMS-user receives the confirmation whether the designated TCC data service has been started or this request has been terminated unsuccessfully.

5.5.7.2 Type of primitives and parameters of DLM-Activate-TCC

Table 15 indicates the primitive and parameters of the DLM-Activate-TCC service.

Table 15 – DLM-Activate-TCC primitives and the parameters

DLM-Activate-TCC	Request	Confirm
Parameter name	input	output
Desired-speed-class	M	
Status		M
NOTE The method by which a confirm primitive is correlated with its corresponding preceding request primitive is a local matter.		

5.5.7.2.1 Desire-speed-class

This parameter specifies the speed class for the TCC data service. The DMLS-user can specify high-speed, medium-speed or low-speed for the speed class.

- a) High-speed
- b) Medium-speed
- c) Low-speed.

5.5.7.2.2 Status

This parameter allows the DLMS-user to determine whether the requested service was provided successfully, or failed for the reason specified. The value conveyed in this parameter is as follows:

- a) "Success – the TCC data service is activated";
- b) "Failure – terminated unsuccessfully".

5.5.8 Deactivate TCC data service

5.5.8.1 Function

This service allows the DLMS-user to deactivate TCC data service. The DLMS-user receives the confirmation whether the designated TCC data service has been terminated or this request has been completed unsuccessfully.

5.5.8.2 Type of primitives and parameters of DLM-Deactivate-TCC

Table 16 indicates the primitive and parameters of the DLM-Deactivate-Publisher service.

Table 16 – DLM-Deactivate-TCC primitives and the parameters

DLM-Deactivate-TCC	Request	Confirm
Parameter name	input	output
Desired-speed-class	M	
Status		M
NOTE The method by which a confirm primitive is correlated with its corresponding preceding request primitive is a local matter.		

5.5.8.2.1 Desire-speed-class

This parameter specifies the speed class for the TCC data service. The DMLS-user can specify High-speed, medium-speed or low-speed for the speed class.

- a) High-speed
- b) Medium-speed
- c) Low-speed.

5.5.8.2.2 Status

This parameter allows the DLMS-user to determine whether the requested service was provided successfully, or failed for the reason specified. The value conveyed in this parameter is as follows:

- a) "Success – the TCC data service is deactivated ";
- b) "Failure — terminated unsuccessfully".

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